Predictive Maintenance for Small Machine Shops

Capstone Project Summary Sandra Hershey-Zagrans · UTD / Fullstack Academy AI/ML Bootcamp · November 2026
Repository: github.com/Alessandrarye/ai4i-capstone

Problem

Small machine shops run equipment until it fails. Predictive maintenance exists, but it is sold to large factories at $15,000 to $40,000 per deployment plus a subscription, and it assumes a reliability engineer on staff. Shop visits conducted as research for this project found good businesses with aging machines and no data about their own equipment. The gap is price, not technology.

Approach

An end-to-end edge platform that runs entirely on one small computer in the shop, with no cloud dependency and no subscription.

- **Data.** AI4I 2020 Predictive Maintenance dataset (Matzka, 2020): 10,000 milling-machine readings, 339 failures (3.4%). Two engineered physics features (temp_delta, power). Failure-mode label columns and identifiers excluded from the feature matrix to prevent leakage. Stratified 75/25 split with a fixed seed.
- **Models.** A gradient-boosted classifier with class weighting and a decision threshold tuned on the precision-recall curve, plus an Isolation Forest anomaly detector trained on healthy readings only to catch failure types the classifier never saw.
- **System.** FastAPI backend scoring each reading with both models, PostgreSQL in Docker Compose, React dashboard showing every machine as a green, yellow, or red tile with KPI cards and trend charts.
- **Live hardware.** An ESP32 with an MPU-6050 accelerometer mounted on a desk fan, posting vibration readings over Wi-Fi every few seconds. Scored against a calibrated vibration baseline (0.07 watch, 0.12 fault). A foam block pressed into the blade path produces a controlled fault for live demonstration.

Results (held-out test set: 2,500 readings, 85 failures)

Metric	Value
Recall, failure class	90.6% (77 of 85 caught)
Precision, failure class	49.7%
F1, failure class	0.642
Threshold	0.046, deliberately low to favor recall
Confusion matrix	2,337 true healthy · 78 false alarms · 8 missed · 77 caught

Accuracy (96.6%) equals a model that always predicts healthy, so it is not used as a success metric. Permutation importance ranks rotational speed and temp_delta highest, then power and tool wear, with raw temperatures near zero, matching the physics of heat-dissipation failures. The Isolation Forest, thresholded at the 2nd percentile of healthy scores, flags about 2% of healthy readings and 15.3% of labeled failures per reading.

Limits

Point-in-time warning, not a countdown (the dataset has no time history). The dataset is synthetic. The rig's vibration feature differs from the dataset's features, so the live path uses a calibrated baseline rather than the fitted models. A fan is not a CNC, and breadboard sensors are not shop sensors. No authentication or staleness detection yet.

Next steps

Return the temperature and current sensors to the rig; fit the Isolation Forest on the rig's own history; mount on a real CNC and learn its baseline; collect time-series data toward lead-time estimates; add authentication and service hardening for a shop pilot.

Reference

Matzka, S. (2020). Explainable artificial intelligence for predictive maintenance applications. *2020 Third International Conference on Artificial Intelligence for Industries (AI4I)*, 69–74. https://doi.org/10.1109/AI4I49448.2020.00023